Blynt — Complete Test Cases Specification

Exhaustive QA Test Gate Flowcharts & Assertion Verifications Across All 14 System Scenarios

Document Overview: This document defines the formal testing architecture for the Blynt accounting platform. Every test suite is presented using a standardized architectural flowchart following the vertical validation gate pattern: passing sequentially through input schema parsing, multi-tenant boundaries, double-entry mathematical balancing, and temporal exclusion constraints, terminating in either green success assertions (HTTP 200/201) or explicit red failure branches (HTTP 400/401/403/409/422).

TC-SEC-001: Multi-Tenant Request Guard & Session Boundary Test

Category: Security & Multi-Tenancy | **Expected Rejection:** ASSERT 401 / 403 / TENANT_MISMATCH

Verification Gate: Intercept unauthenticated or cross-tenant API requests

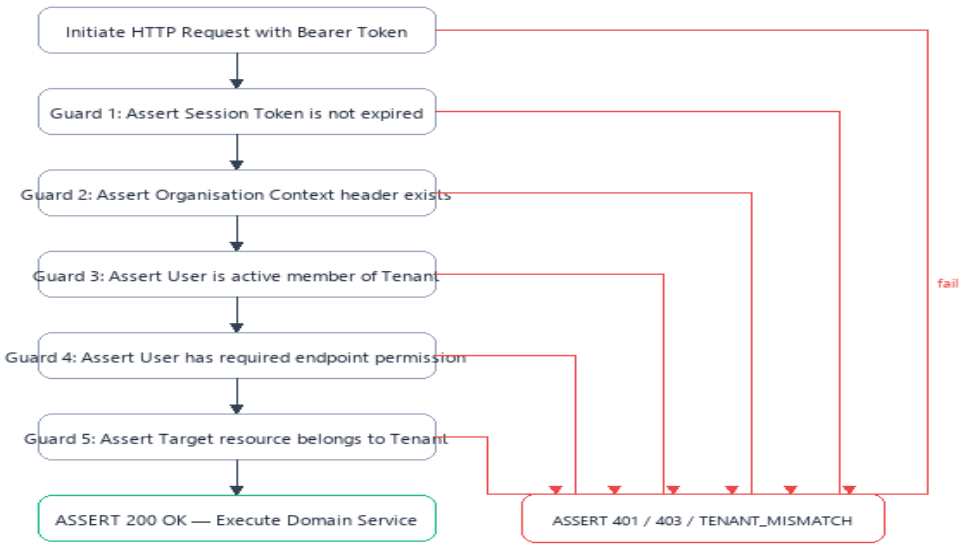


Figure TC-SEC-001 — Multi-Tenant Request Guard & Session Boundary Test QA Test Gate Architecture

Assertion Rule: Requests with forged tokens or mismatched organization IDs must be rejected immediately before invoking business logic.

TC-ORG-001: Organization Creation & Unique Slug / VAT Validation

Category: Tenancy & Setup | **Expected Rejection:** ASSERT 400 / 409 / SLUG_OR_VAT_INVALID

Verification Gate: Validate business entity registration parameters

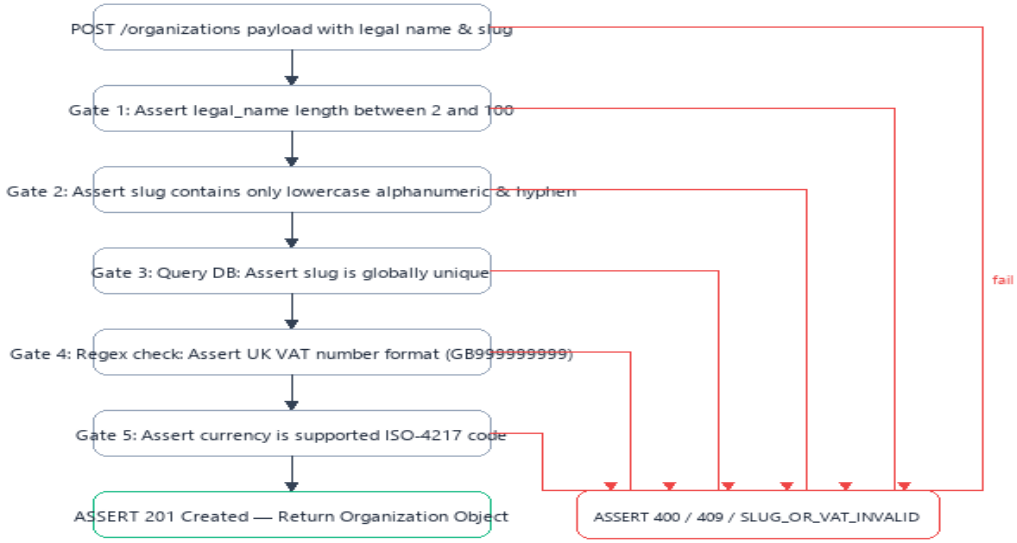


Figure TC-ORG-001 — Organization Creation & Unique Slug / VAT Validation QA Test Gate Architecture

Assertion Rule: Duplicate company slugs or invalid HMRC VAT formats must be rejected with HTTP 409 and 400.

TC-COA-001: Chart of Accounts Control Immutability & Hierarchy Rules

Category: General Ledger Governance | **Expected Rejection:** ASSERT 400 / 422 / CONTROL_ACCOUNT_PROTECTED

Verification Gate: Protect control accounts from manual tampering

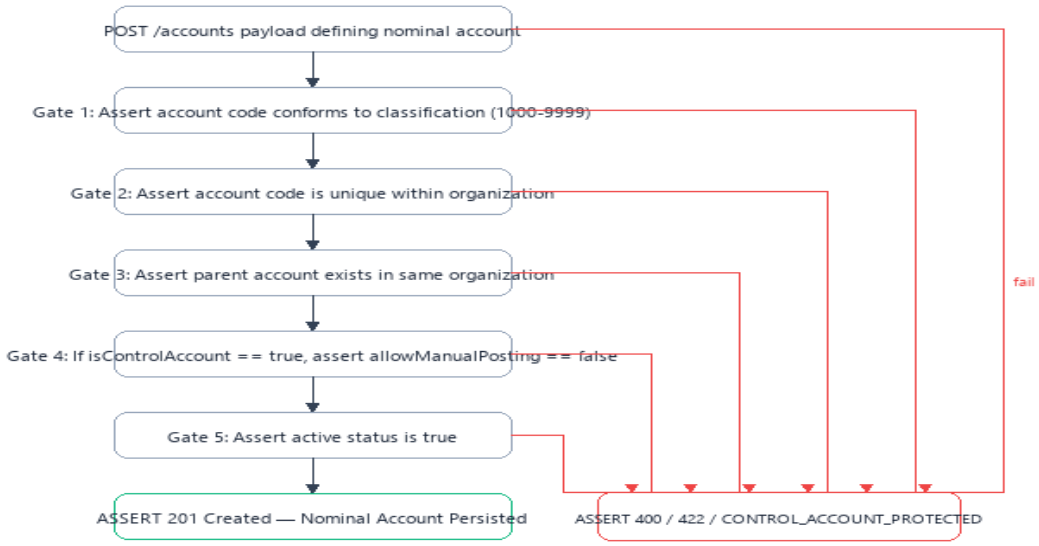


Figure TC-COA-001 — Chart of Accounts Control Immutability & Hierarchy Rules QA Test Gate Architecture

Assertion Rule: Control accounts cannot have manual posting enabled; duplicate codes within the same tenant must be rejected.

TC-INV-001: Sales Invoice 20% VAT Calculation & Subledger Sync

Category: Sales & Invoicing | **Expected Rejection:** ASSERT 400 / 422 / INVOICE_VALIDATION_FAILED

Verification Gate: Verify arithmetic precision and synchronous posting

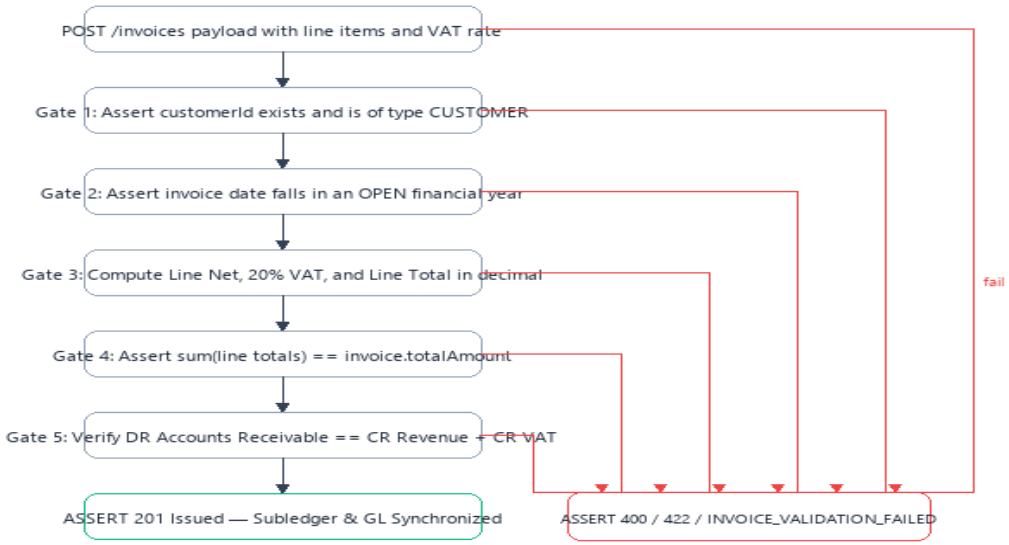


Figure TC-INV-001 — Sales Invoice 20% VAT Calculation & Subledger Sync QA Test Gate Architecture

Assertion Rule: Total amount must equal exact sum of lines plus 20% VAT; customer balance must increase by totalAmount immediately.

TC-PAY-001: Payment Gateway Idempotency & Double-Click Shield

Category: Client-Side & Payments | **Expected Rejection:** ASSERT 400 / 402 / CARD_DECLINED_OR_DUPLICATE

Verification Gate: Prevent duplicate charges on rapid successive clicks

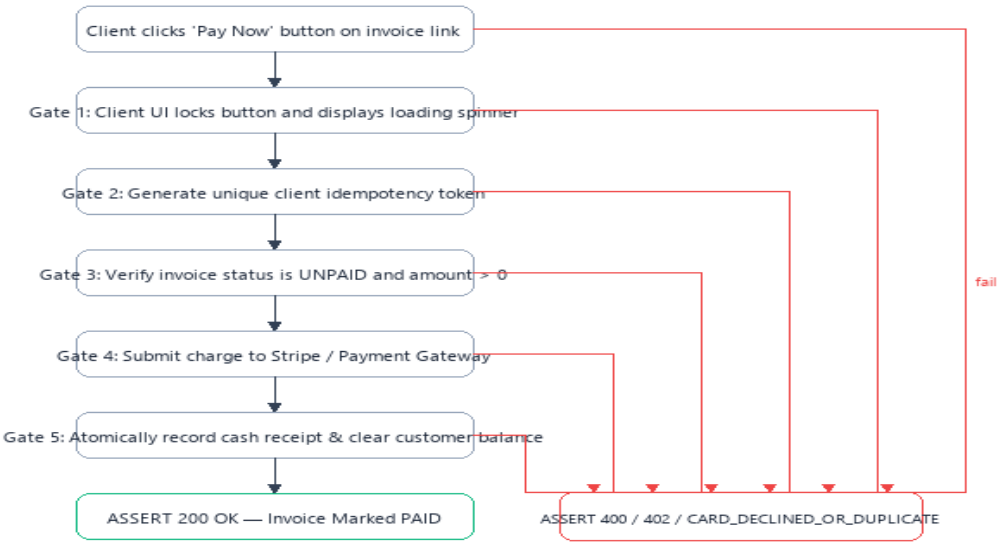


Figure TC-PAY-001 — Payment Gateway Idempotency & Double-Click Shield QA Test Gate Architecture

Assertion Rule: Successive clicks within 5 seconds must produce zero duplicate charges; invoice status must transition atomically to PAID.

TC-PUR-001: Supplier Bill Duplicate Interception & AP Posting

Category: Purchases & Accounts Payable | **Expected Rejection:** ASSERT 409 / DUPLICATE_BILL_REFERENCE

Verification Gate: Block duplicate supplier reference bills

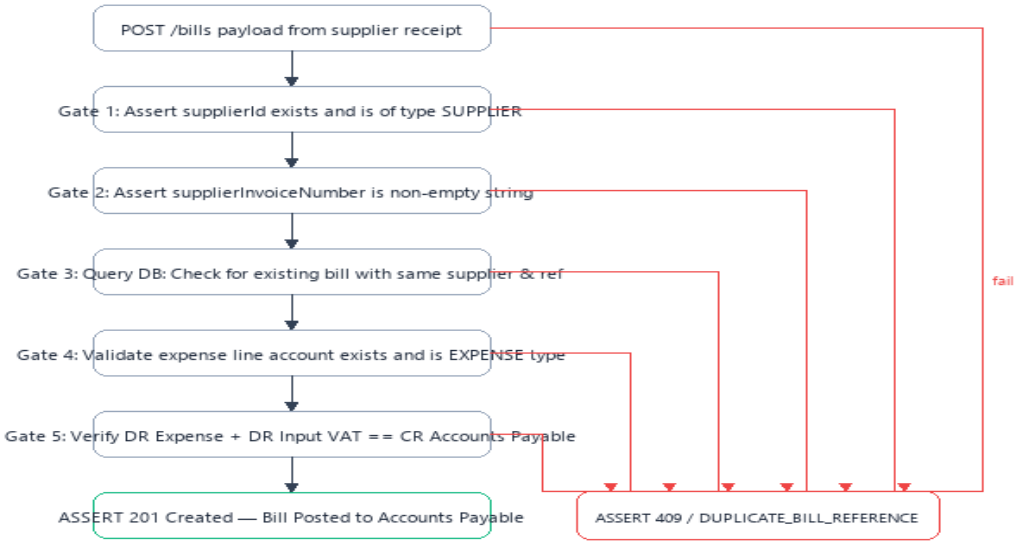


Figure TC-PUR-001 — Supplier Bill Duplicate Interception & AP Posting QA Test Gate Architecture

Assertion Rule: Attempting to enter a supplier bill with an existing reference for that supplier must fail with HTTP 409.

TC-AI-001: AI Document OCR Confidence & Arithmetic Guardrails

Category: AI Intelligence & Safety | **Expected Rejection:** ASSERT 422 / AI_LOW_CONFIDENCE_REJECTED

Verification Gate: Block hallucinated OCR extractions

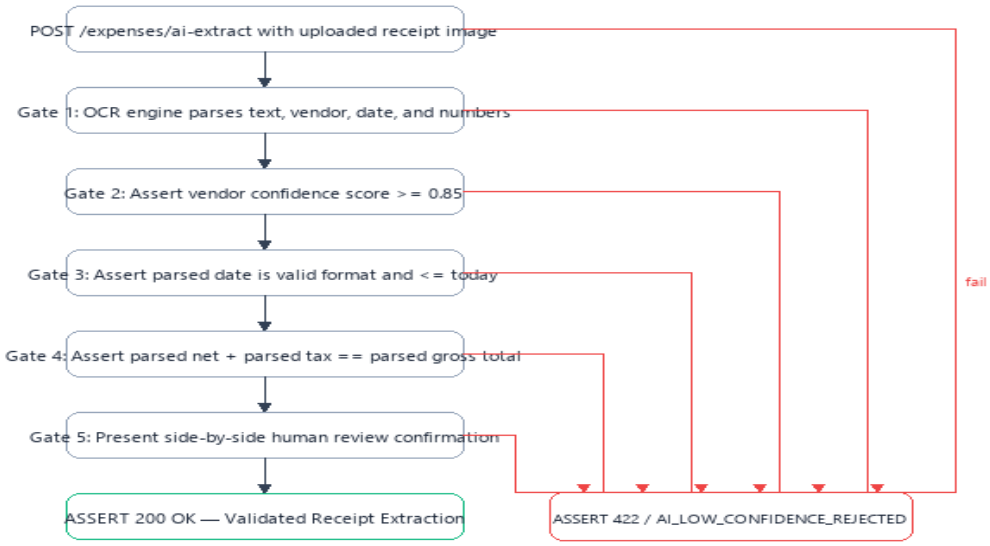


Figure TC-AI-001 — AI Document OCR Confidence & Arithmetic Guardrails QA Test Gate Architecture

Assertion Rule: If confidence is below 85% or line math does not equal total, automatic posting is blocked; human review is required.

TC-JRN-001: Journal Balance Equilibrium Invariant (Debits == Credits)

Category: General Ledger Engine | **Expected Rejection:** ASSERT 422 / UNBALANCED_JOURNAL_ENTRY

Verification Gate: Enforce double-entry accounting equilibrium

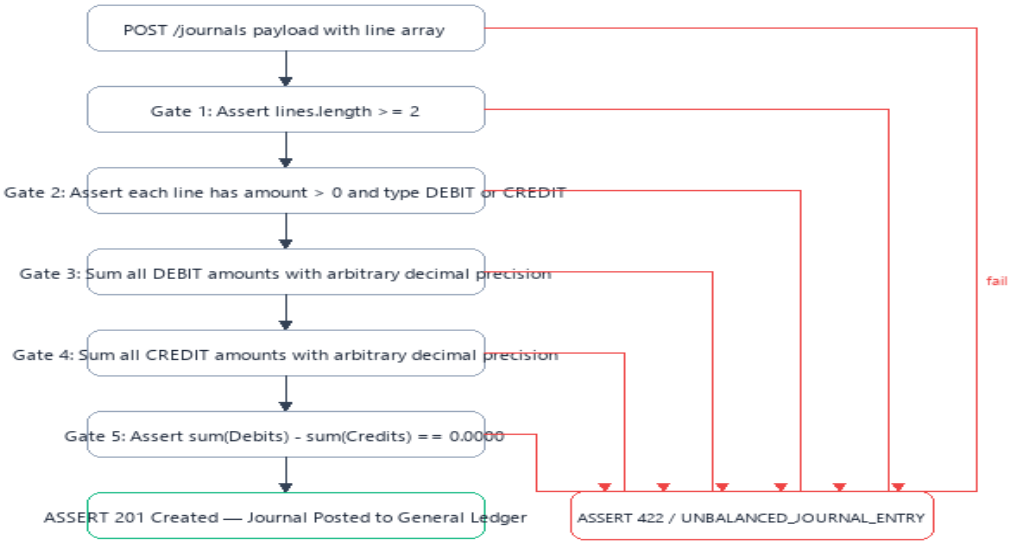


Figure TC-JRN-001 — Journal Balance Equilibrium Invariant (Debits == Credits) QA Test Gate Architecture

Assertion Rule: Any journal entry with a non-zero difference between debits and credits must be rejected with HTTP 422.

TC-REV-001: Invoice Voiding Reversal Invariant & Customer Zeroing

Category: GAAP Reversals & Audit | **Expected Rejection:** ASSERT 400 / 409 / CANNOT_VOID_PAID_INVOICE

Verification Gate: Verify symmetric non-destructive cancellation

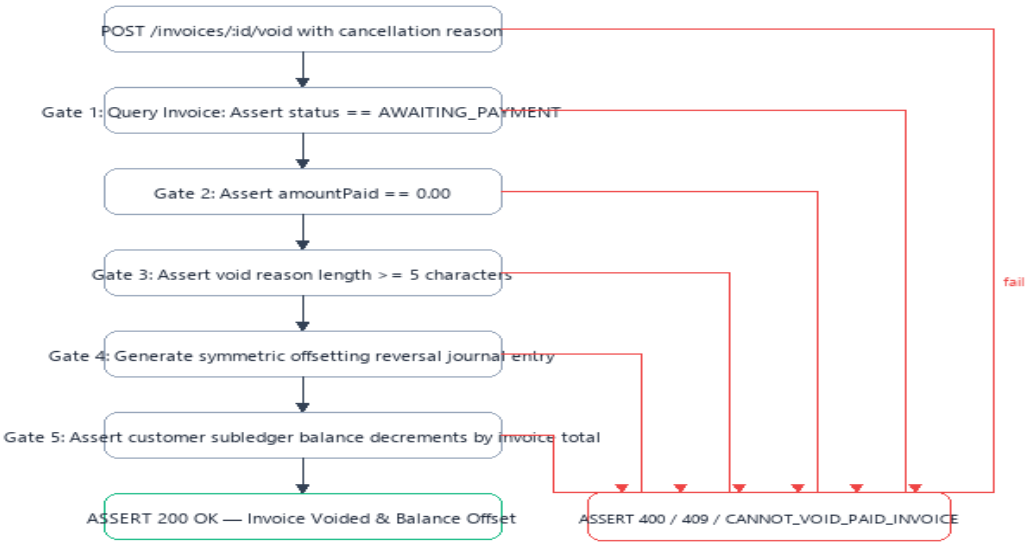


Figure TC-REV-001 — Invoice Voiding Reversal Invariant & Customer Zeroing QA Test Gate Architecture

Assertion Rule: Voiding must create a mirrored reversal journal, customer balance must return to pre-invoice state, and original invoice is preserved.

TC-PER-001: Closed Financial Period Temporal Backdating Intercept

Category: Period Governance | **Expected Rejection:** ASSERT 403 / FINANCIAL_PERIOD_LOCKED

Verification Gate: Block transactions backdated into closed tax years

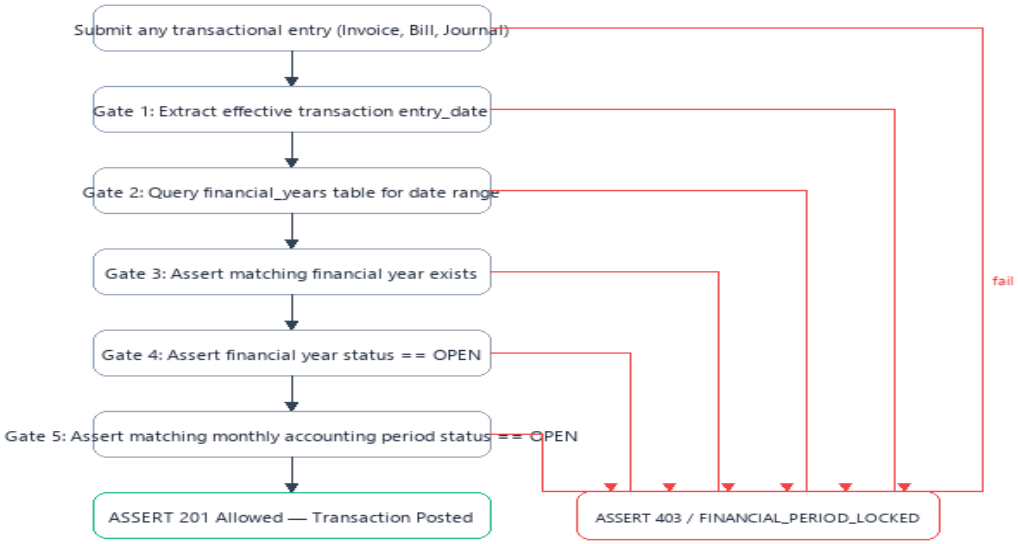


Figure TC-PER-001 — Closed Financial Period Temporal Backdating Intercept QA Test Gate Architecture

Assertion Rule: Attempting to post any entry with an entry_date inside a closed or locked financial year must fail with HTTP 403 PERIOD_LOCKED.

TC-BNK-001: Bank Feed Statement Deduplication & Reconciliation

Category: Banking & Reconciliation | **Expected Rejection:** ASSERT 409 / DUPLICATE_TRANSACTION_FEED

Verification Gate: Prevent duplicate bank feeds and ensure balance matching

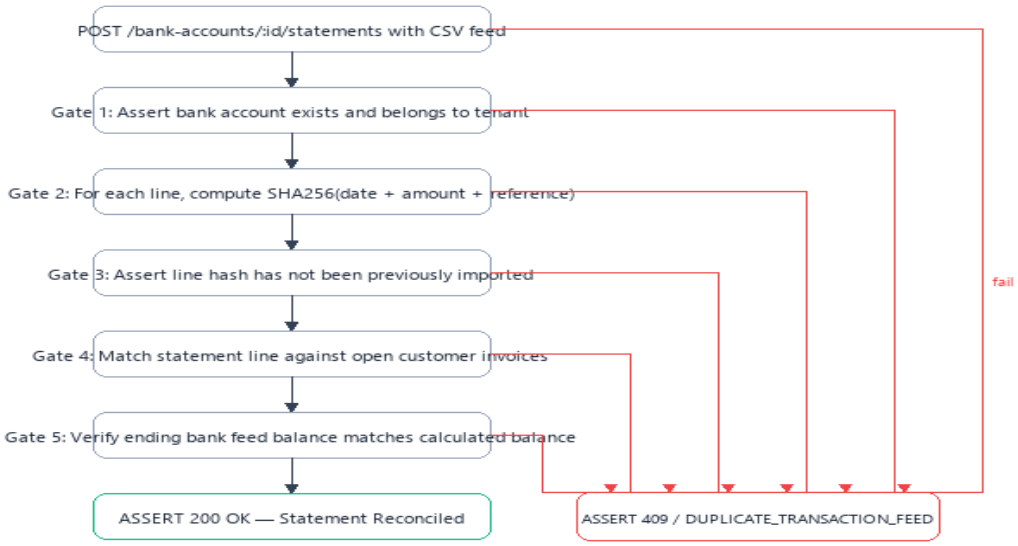


Figure TC-BNK-001 — Bank Feed Statement Deduplication & Reconciliation QA Test Gate Architecture

Assertion Rule: Duplicate imported statement lines must be ignored or rejected; clearing an invoice must update both bank and AR subledgers.

TC-RBAC-001: Role Privilege Boundary Enforcement (Viewer vs Admin)

Category: Access Control Security | **Expected Rejection:** ASSERT 403 / INSUFFICIENT_PERMISSIONS

Verification Gate: Verify write operation prohibition for Viewer role

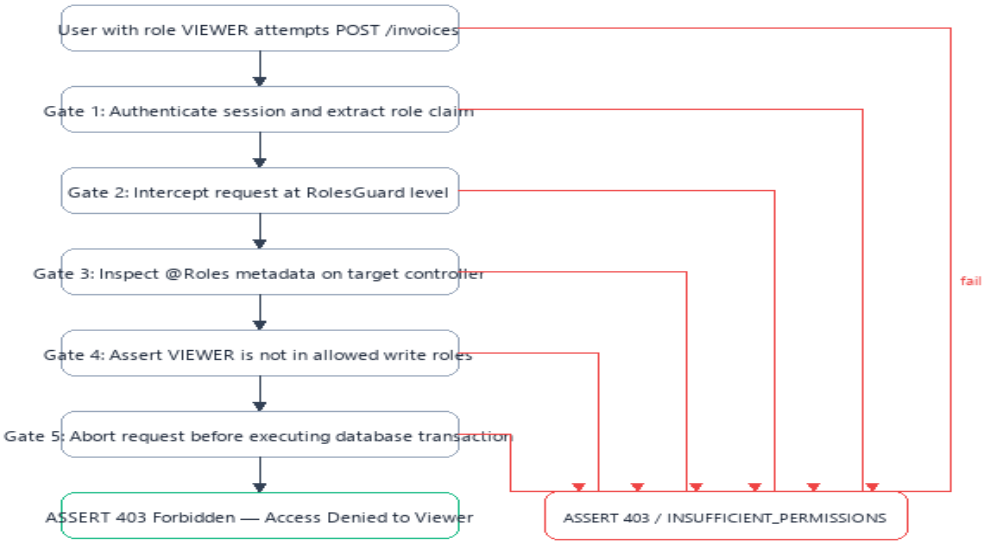


Figure TC-RBAC-001 — Role Privilege Boundary Enforcement (Viewer vs Admin) QA Test Gate Architecture

Assertion Rule: Users with the VIEWER role must receive HTTP 403 when attempting any POST, PUT, PATCH, or DELETE operation.

TC-VAT-001: Quarterly VAT Return Derivation & HMRC MTD Validation

Category: Statutory Compliance | **Expected Rejection:** ASSERT 422 / VAT_MATHEMATICAL_DISCREPANCY

Verification Gate: Assert 9-box VAT mathematical equality

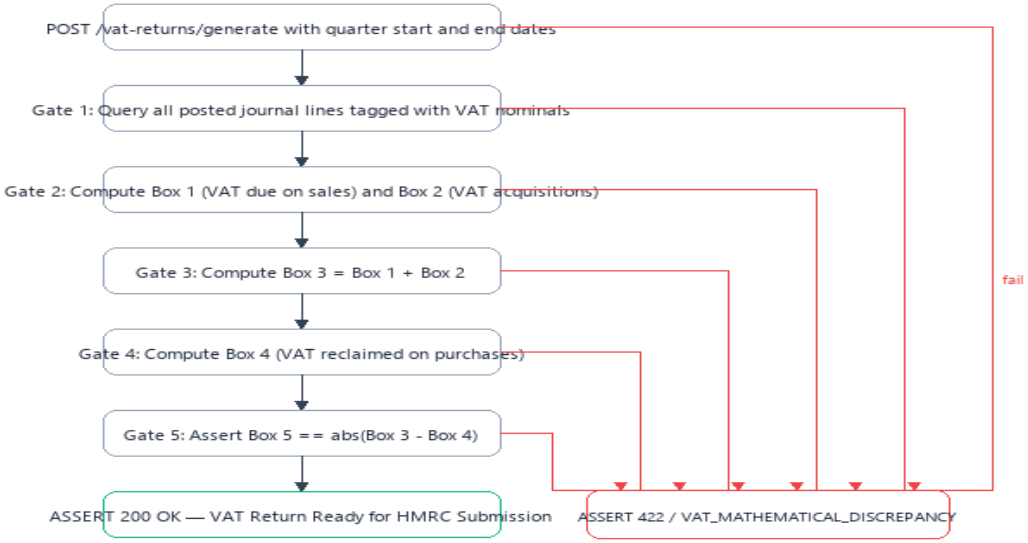


Figure TC-VAT-001 — Quarterly VAT Return Derivation & HMRC MTD Validation QA Test Gate Architecture

Assertion Rule: Box 5 must equal exactly Box 3 minus Box 4; no unposted or draft transactions may be included in the statutory calculation.

TC-AUD-001: Audit Trail Append-Only Immutability & Hash Verification

Category: Statutory Audit & GRC | **Expected Rejection:** ASSERT 500 / AUDIT_IMMUTABILITY_VIOLATION

Verification Gate: Verify cryptographic tamper-proofing of audit records

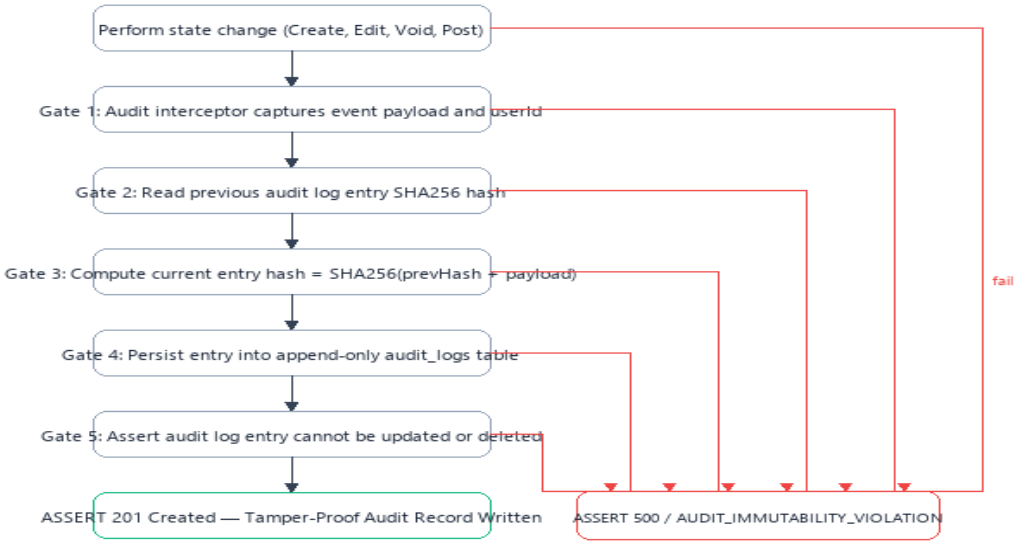


Figure TC-AUD-001 — Audit Trail Append-Only Immutability & Hash Verification QA Test Gate Architecture

Assertion Rule: Direct SQL UPDATE or DELETE on audit_logs must be rejected by PostgreSQL triggers; hash chain continuity must verify 100% intact.